



K-7441

**Integrated M.C.A. (Sem.-V)
Examination - December-2024**

IMCA-501T

Python Programming

Time allowed : Three Hours

Maximum Marks : 70

Min. Pass Marks : 28

*Note : This question paper contains two sections
as under :*

Section-A**Max. Marks-20**

This section will have 10 questions, having 02 questions from each unit with no internal choice. Answer in about 20 words. All questions will have equal marks. (10×02=20)

Section-B**Max. Marks-50**

This section will have 10 questions, having 02 questions from each unit, one question to be attempted from each unit. Word limit approximately 400-500 words. Questions may have sub divisions. All questions will have equal marks. (10×05=50)

SECTION - A

- 1 (i) What is the difference between mutable and immutable data types?
- (ii) What is block of code in Python?
- (iii) What is the purpose of break, continue and pass statement in Python?
- (iv) What is the difference between iterating over a list and a dictionary using a for loop?
- (v) Explain list slicing with an example.
- (vi) What is a tuple in Python? How is it different from a list?
- (vii) How do you open a file in Python? Explain different file modes.
- (viii) Explain the use of the seek() and tell() functions in file handling.
- (ix) What is Python package? How is it different from a module?
- (x) What is Tkinter and why is it used in Python?

K-7441]**2****[Contd...****SECTION - B****UNIT - I**

- 2/ What are operators in Python? Explain different types of operators with examples.

OR

Explain different data types in Python with examples. Explain type conversion in Python with example.

UNIT - II

- 3 What is the range() function in Python? Explain with examples, write a Python program to print the sum of squares of first N natural numbers using a for loop.

OR

Explain the working of conditional statements in Python. Discuss their importance in decision making. Write a Python program that takes three numbers as input and determines if they form a valid triangle.

UNIT - III

- 4/ What are dictionaries in Python? Explain various operations performed on dictionaries with example. Write a Python program to merge two dictionaries and sort them based on keys.

OR**K-7441]****3****[Contd...**

What are different string operations available in Python? Write a Python program that takes a string as input and counts the number of uppercase, lowercase, digits and special character in it.

UNIT - IV

5 What are the different type of file operations in Python? Explain file reading and writing methods with examples.

OR

Compare text files and binary files in Python. How do operations differ when working with each type? Provide example to illustrate your answer.

UNIT - V

6 Discuss the features of Pandas. Explain how Pandas can be used for data cleaning, data transformation and data analysis with suitable examples.

OR

Explain the concept of Tkinter widgets. Discuss at least five important widgets with their functions and provide a small example of how they are used.
